Prior Audit–Repair Context Shifts LLM Verifier Thresholds Toward Leniency

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Abstract

Automated checking pipelines increasingly place one language model as the checker and another (or the same one) as the fixer. We ask whether that wiring changes what the checker reports. Measuring false alarms on human-verified-correct ProcessBench traces with the present task held byte-identical, we find that a completed audit $\rightarrow$ repair episode already in the model's context lowers false alarms in **15 of 15** model $\times$ wording combinations, by 2.8 to 11.5 pp against a length-matched non-audit control, a 9 to 25% reduction relative to that control. The direction contradicts what the accumulated-message literature predicts: an episode whose audit reported an *error* lowers false alarms *further* still, at **all five** wordings on the model where that manipulation lands cleanly, though a negativity asymmetry predicts more flagging. Decomposing the episode finds repair content and audit verdict complementary: different components carry the effect on different model families. Signal-detection analysis locates the change in the threshold rather than in discrimination — the criterion moves in 15 of 15 combinations and survives correction in 13 while d' survives in none, though the d' test is half as sensitive by construction — and a hand audit of 50 false alarms finds 82% simply wrong, so at this operating point the shift need not be harmful. With reasoning enabled the effect keeps its relative size on both models tested, and the threshold reading holds there too.

1 Introduction

When a language model checks work — reviewing code, grading a proof, verifying a reasoning trace — it is increasingly wired to something downstream: sometimes a second model repairs what it flags, sometimes it is handed its own findings and asked to fix them. Choosing between those is treated as an engineering detail, a question of how to arrange the pipeline rather than of what the checker will say.[1]

This paper shows it is not a detail, and that the direction of the effect is not the one the nearest literature predicts. The practical finding is a caution: **be wary of placing a verifier in a context where it has already carried out the kind of repair it is about to be asked to judge**. Its false-alarm rate falls by up to 11.5 pp with no gain in discrimination we can detect, and a metric computed on flagged items alone will read that as an improvement.

Two recent results point opposite ways, and neither isolates the manipulation. Jin and Chen (2026) find that prompt formats demanding explanations and fixes *increase* misjudgement of correct code, but there the fix is requested in the same response as the audit, so the effect is confounded with output format. Khullar et al. (2026) find monitors going easier on work framed as their own, yet report that "explicitly stating that the action comes from the monitor does not by itself induce self-attribution bias": their effect rides on *where* the work sits, not whose it is said to be.

Our design separates what those conflate. The present task (problem text, step text, instructions, output schema, evidence budget) is **byte-identical** across conditions, enforced by tests that diff rendered prompts, so nothing here is an output-format effect; and the audit $\rightarrow$ repair context is a completed exchange about a *different* item, never a rewording of the request being answered.

The dependent variable is the **false-alarm rate** (FAR): how often a model reports an error in a trace human annotators verified as correct. FAR rather than accuracy, because the manipulation is hypothesised to move a *threshold*, and because each false alarm costs a repair cycle on something already right.

Contributions.

[1] Code: https://github.com/parsa-mz/crtitxer

1. **The effect** (§4). A prior audit → repair episode lowers false alarms in 15 of 15 model × wording combinations against a length-matched filler, surviving five controls.
2. **Polarity drift alone cannot explain it** (§5). An episode reporting an *error* lowers false alarms *further*, at all five wordings on the model where that cell is clean, which is the opposite sign to what the accumulated-message account predicts. Decomposing further, repair content and the audit's verdict are complementary across models, so no single component is necessary.
3. **What it does to the instrument** (§6). The change is a criterion shift with no discrimination gain we can detect, and a hand audit of the false alarms it removes shows the shift is nonetheless beneficial at this operating point.

2 Related work

Prior conversation shifts judgements the other way. The closest result is Temkit (2026), who across 84,088 calls to 12 models shows that judgements drift toward the polarity of the preceding conversation ($d = -0.17$), concentrated on items where the model is uncertain at baseline, with a **negativity asymmetry**: negative histories induce $1.52\times$ more drift than positive ones. That account makes a sharp prediction here. Our incorrect-verdict cell places an audit reporting an *error* in the context (an unambiguously negative history) and so should raise the false-alarm rate. It lowers it, and by more than a clean audit does (§5). That work also reports the drift neither growing with context length (5 prior turns and 50 give the same shift) nor varying with position, so neither is available to explain a *larger* effect.

Verifier strictness as a manipulable quantity. Zhou et al. (2026) steer verifier strictness directly on ProcessBench, the same benchmark, establishing it as a movable axis. What they do not ask is whether an ordinary pipeline arrangement moves it with nobody intending to.

Judges, and what makes them move. LLM-as-judge evaluation is now standard (Zheng et al., 2023; Gu et al., 2024), with documented sensitivity to self-preference (Panickssery et al., 2024), to sycophantic agreement with a user (Sharma et al., 2023), to the ordering of in-context examples (Lu et al., 2022), to provided knowledge (Li et al., 2025) and to paraphrase (Bellibatlu et al., 2026). Ours is a different lever: not who the judge is talking to or how the question is worded, but what job the judge has been told comes next.

The critique → correction pipeline itself is benchmarked by CriticBench (Lin et al., 2024) and CriticEval (Lan et al., 2024), and Yang et al. (2025) decompose self-correction into confidence and critique components; self-correction work more broadly asks whether models can repair their own output (Huang et al., 2024; Olausson et al., 2024). All of these measure how well the critique and the correction are done. We ask something upstream of that: what a completed repair already in the context does to the critique itself.

Signal detection. A false-alarm rate on its own cannot separate a model that discriminates better from one that has become reluctant to flag. We report d' and the criterion c (Macmillan and Creelman, 2004), following recent applications of signal-detection theory to language models (Cacioli, 2026), and treat the pairing of a false-alarm rate with a detection rate on labelled-incorrect traces as the minimum needed to interpret any FAR movement at all.

3 Method

3.1 Task and dependent variable

Each item is a mathematical problem with a step-by-step candidate solution. The model returns one JSON object under a constrained decoding schema: verdict, first-error step, confidence, error type, and at most forty words of evidence (Appendix E). Reasoning traces are disabled deliberately, because variable-length thinking would break the identical-budget invariant the comparison rests on; §7 turns them back on.

Items come from ProcessBench (Zheng et al., 2024), using only traces whose label marks every step correct, so *a reported error is a false alarm by construction*. From an eligible pool of 1,101 we allocate three disjoint, source-stratified arms: **929 clean** targets, **50 warmup** items for generating episodes, and 122 held in reserve. Disjointness is load-bearing: an episode item that was also a target would mean the model had already audited the very trace it is later asked to judge cold.

The signal side of the signal-detection analysis (§6) is a second, disjoint arm of **929 labelled-incorrect** traces drawn with the same seed and matched to the clean arm's source mix trace for

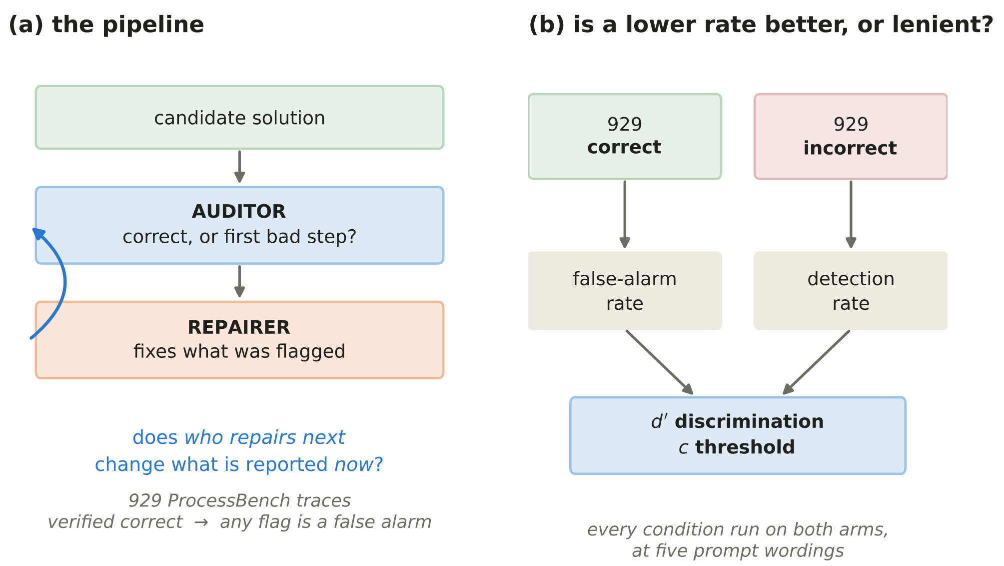

Figure 1: The question and the instrument. **(a)** An auditing pipeline: a checker reports, a repairer fixes. We ask whether *who repairs next* changes what is reported *now*, on traces human annotators verified correct, so any flag is a false alarm. **(b)** A false-alarm rate alone cannot distinguish better discrimination from greater reluctance to flag, so every condition is run on both a correct and a labelled-incorrect arm and combined into d' and the criterion c. The two arms are disjoint and matched trace for trace on source; the prior-context conditions use a source-proportional half of the correct arm (465 targets), the ladder all 929.

trace (126 GSM8K, 315 MATH, 301 OlympiadBench, 187 OmniMath); the two arms share no item. The incorrect-verdict episodes (AX, AXN) come from a further 50 labelled-incorrect traces per model, source-stratified and verified to overlap neither the 929 detection targets nor the 1,101-trace correct pool. "Detection rate" throughout means the rate at which a model returns the `incorrect` verdict on those traces, not whether it localises the faulty step.

FAR is measured from **8 samples per item at** $T{=}0.7$. Sampling is required, not incidental: at $T{=}0$ each item contributes a hard $0/1$, so a manipulation that shifts every item's report propensity a few points while flipping few argmaxes would be invisible.

We audit three models (Qwen3.6-27B, Qwen3.6-35B-A3B and Ministral-3-14B), chosen by a pre-specified instrument-sensitivity screen that a further two candidates failed (Appendix A): it asks whether a model's FAR responds to explicit framing *at all*, in either direction, since a dependent variable that does not move cannot inform a null. Every condition is measured under **five semantically matched wordings** (F1–F5) of the audit instruction (§7).

3.2 The prior-context conditions

The manipulation places a completed audit → repair episode on a held-out item into the context before the target request, on a source-proportional half of the clean arm (465 of 929 targets), with the no-context baseline **R0** re-measured inside that subset. Episodes are generated by each auditor itself at $T{=}0$ and then frozen, so the episode is genuinely that model's own work and is byte-identical across cells.

Attributing an episode to another model requires either relabelling an assistant turn (a *stated* attribution, which Khullar et al. (2026) report is null) or moving it into a user turn, which reintroduces their placement confound. Neither alone is interpretable, so we cross them into a 2×2: **AS** (assistant turn, self), **AO** (assistant, labelled peer), **US** (user, labelled self), **UO** (user, peer). Labels are length-matched, and tests enforce that stripping them makes all four cells textually identical. Throughout, "episode" in a contrast means the **mean of those four cells**; a contrast naming AS means AS.

3.3 Five controls

The 2×2 was close to flat on the two Qwen models (all four cells within 0.63 and 0.72 pp of each other) but not on Ministral, whose cells span 5.65 pp because that is the one model with a placement effect (Appendix B). What all four share and R0 lacks is *a prior exchange at all*, which the 2×2 cannot distinguish from the episode's content. Each control below removes one more alternative.

AF, length-matched filler. A prior exchange on a non-audit task, in the same turns, paired to the same targets, length-matched *per episode* (Appendix E). Separates "the episode was an audit" from "there was context".

AV, audit only. The episode with the repair request and repair deleted. AV is *shorter* than AF, so AS $-$ AV varies whether a continuation exists as well as what it contains, which is why we rest nothing on it.

AN, inert continuation. The repair request replaced by "restate, without changing it, what your audit concluded", and the repair by the model's own restatement, generated in the episode's own context at $T{=}0$. AS and AN both carry an audit *and* a second assistant turn, differing only in what that turn does, which is what makes AS $-$ AN identified where AS $-$ AV is not. The restatements came out 20–35% *longer* in mean characters than the repairs they replace, so the contrast is conservative with respect to length.

AX, incorrect verdict. The same structure on a ProcessBench trace *labelled incorrect*, so the model reports an error and produces a real correction. This control exists because of an asymmetry in the episodes: on the Qwen models 88% and 86% of ordinary episodes report `correct` (Ministral, 34%, is the exception), and a clean verdict leaves nothing to correct, so those cells place one or two `"verdict": "correct"` assertions in front of a model about to emit that field. AF cannot separate that from repair content: it deletes verdict and repair together.

AXN, inert continuation on the fault. AN's construction on AX's pool. This makes AX $-$ AXN the only contrast where the thing removed is a *real repair of a real fault*. AXN $-$ AN then compares an error-verdict episode with a clean-verdict one having removed the repair from both sides. Note what it does *not* isolate: the two sides draw on different pools, so the episode's trace varies with its verdict, and separating the two would need a fourth pool we have not run.

The resulting decomposition, with what each contrast holds fixed:

contrast	what varies
episode $-$ AF	audit content, length held
AV $-$ AF	audit alone vs non-audit
AN $-$ AF	audit + inert turn vs non-audit
AS $-$ AN	continuation content only
AX $-$ AXN	content only, on a real fault
AXN $-$ AN	error vs clean episode, repair removed from both
AX $-$ AS	polarity *and* repair reality
AS $-$ AV	presence *and* content (not identified)

Under polarity drift (Temkit, 2026), an episode whose audit reports an error should push FAR *up* relative to one reporting none. Under experienced repair it should behave like AS. The two make opposite directional predictions, which is what makes AX decisive.

3.4 Statistical conventions

Three choices, each load-bearing.

Intervals cluster on the reused episode. All intervals are 20,000-replicate bootstraps on per-item paired differences. Each prior-context cell cycles 465 targets over a pool of 50 frozen episodes, so **each episode is reused for about 9.3 targets** and two targets sharing one see the same prior exchange. Every such interval is therefore a cluster bootstrap over episodes, which puts the effective sample size for the episode-attributable component nearer 50 than 465, widening the headline contrast's interval by a median of 14% against item resampling (over all claim contrasts 7%, and narrower in about a quarter; both are persisted). Calibration was verified by simulation at this study's geometry (coverage 0.94 against a nominal 0.95). d', c and balanced accuracy pair a false-alarm rate on 465 targets with a detection rate on 929, so no per-item pairing exists across the two: each arm is resampled independently and clustered on its own episode assignment (9.3 targets per episode on the clean side, 18.6 on the incorrect side), and the two draws are combined per replicate.

One significance criterion. Every resampled p is the achieved significance level of the interval printed beside it, so the two cannot disagree. The hand audit alone is not resampled; its exact binomial intervals and Fisher test are named where they appear.

Multiplicity over families declared in code. Holm–Bonferroni step-down within families fixed before any p-value was visible, since a family chosen afterwards is not a correction. The claim family is the eight contrasts of §3.3 across three models,

$k = 24$; the 2×2 factorial contrasts form their own family, $k = 9$; and $\Delta d'$, Δc and Δbalanced accuracy for the three prior-context cells (AF, AS, AV) across three models form one family each, $k = 9$. The last two matter: the paper concludes from criterion counts as well as from d' counts. Each wording is corrected as its own set of families over the same declared membership, so cross-wording summaries are counts, never p-values. † marks survival of the declared family and is the only mark we draw a conclusion from.

4 A prior audit–repair episode lowers false alarms

Against the length-matched filler the effect is -4.00, -3.59 and -8.83 pp at our primary wording (Table 1), and it holds in **all 15** model × wording combinations, ranging -2.8 to -11.5 pp, every p at the 20,000-replicate floor after episode clustering. This is the result that survives every control we applied.

The filler is what makes it a claim about audits rather than about context. Taking the models in Table 1's column order, AF − R0 is $+1.58$, -0.18 and -0.30 pp ($p = 0.0045$, 0.86, 0.71): null on two of three, and on Qwen3.6-27B it moves *opposite* to the episode, which makes that model's episode − R0 figure conservative rather than inflated. Prior context of any kind does not do this; prior context that was an audit does.

Two further readings from Table 1, and both need the full sweep (Table 4) rather than F1 alone. At F1 the repair *request* survives on **all three** models — AS − AN is -1.31, -2.35 and -3.13 pp, every one clearing $k = 24$ — while the audit with an inert turn in place of the repair (AN − AF) survives on **one**. Across all five wordings that ordering reverses: AN − AF survives **12 of 15** and AS − AN **10 of 15**. So neither component is dispensable and neither is safe to read at one wording, which is the general point of §7. Merely having a continuation contributes on none (AV − AN: -0.10, -0.69, -0.63 pp, all $p > 0.2$).

It also cuts against a simple instruction-following account. On Qwen3.6-35B-A3B the episode moves FAR by -3.59 pp while a *maximal explicit leniency prime* on that model moves it only -1.76 pp $[-3.72, +0.07]$, an interval touching zero, so not an established effect at all (Appendix A): whatever a prior audit does to this model, an explicit instruction does not reach.

5 Polarity drift alone cannot explain it

5.1 The decisive contrast has the wrong sign

If a prior audit lowers false alarms because the model drifts toward the polarity of its context, then an audit reporting an *error* (the negative pole, and the one Temkit (2026) finds $1.52\times$ stronger) must raise them. It does the opposite. On Ministral-3-14B, AX − AS is -5.62 pp $[-8.27, -2.87]$ at our primary wording and negative at **all five**, ranging -4.05 to -5.90 pp with every one surviving its declared family (Table 2). An episode in which the model found and fixed a real error lowers subsequent false alarms *more* than one in which it found nothing: the sign opposite to the prediction, at all five wordings. Against R0 the faulty-episode cell moves FAR by -3.27, -7.30 and -12.09 pp across the three models. A within-pool split of Ministral's own episodes by the verdict they reached converges with this and needs no extra cell (Appendix D).

The other two models do not carry the test, and the reason is measurable rather than mysterious. The manipulation only lands if the faulty episode actually contains a correction, and on Qwen3.6-27B **37 of 50** of those "repairs" still end on a `"verdict": "correct"` assertion; its contrast is null at every wording. On Qwen3.6-35B-A3B the effect is large at F1 (-3.83 pp $[-5.51, -2.23]$) but does not reach its family at the other four, so we count the wedge as one model of three rather than reading a single wording.

What AX − AS rules out is polarity drift and verdict echoing. It does not rule out broader semantic priming or base-rate calibration, because AX varies content and difficulty as well as verdict. The content-matched verdict flip that would settle it needs a different design rather than more compute: forcing the verdict would break the invariant that episodes are the model's own work.

5.2 No single component is necessary

Three components separate: what the repair *content* adds (AX − AXN, both sides carrying a genuine `incorrect` audit), what an *error-verdict episode* does once the repair is removed from both sides (AXN − AN), and what the repair *request* elicits (AS − AN). Measured at all five wordings (Table 2), they are **complementary rather than consistent**:

- Repair content carries the two Qwen models, surviving at **4 of 5** wordings each, and is null on

	Qwen3.6-27B	Qwen3.6-35B-A3B	Ministral-3-14B
R0 false-alarm rate	0.185	0.232	0.691
Is it an audit, or merely context?			
episode − AF	$-4.00^\dagger$ [−5.26, −2.80]	$-3.59^\dagger$ [−5.38, −1.92]	$-8.83^\dagger$ [−11.83, −5.91]
AF − R0 (the filler itself)	+1.58 [+0.46, +2.71]	−0.18 [−2.12, +1.69]	−0.30 [−1.83, +1.18]
Which part of the episode?			
AV − AF (audit alone)	$-3.13^\dagger$ [−4.38, −1.98]	−1.62 [−3.48, +0.19]	−3.67 [−6.39, −0.96]
AN − AF (audit + inert turn)	$-3.03^\dagger$ [−4.27, −1.85]	−0.93 [−2.72, +0.79]	−3.04 [−5.97, −0.14]
AS − AN (repair request)	$-1.31^\dagger$ [−1.97, −0.73]	$-2.35^\dagger$ [−3.81, −0.88]	$-3.13^\dagger$ [−5.41, −0.90]
AV − AN (continuation presence)	−0.10 [−0.91, +0.89]	−0.69 [−1.85, +0.42]	−0.63 [−2.56, +1.43]

Table 1: The episode effect and its controls at wording F1, 465 targets, FAR percentage points; negative means *fewer* false alarms. Every interval is a 95% cluster bootstrap on the frozen episode (§3.4). "episode" is the mean of the four 2×2 cells, not AS alone. † marks survival of Holm–Bonferroni within the declared claim family ($k = 24$), the only mark we draw a conclusion from. Ministral's AV − AF and AN − AF are therefore shown with intervals excluding zero that we nonetheless do not claim.

Ministral at all five.

- The error-verdict episode is the mirror image: **5 of 5** on Ministral, where it is the largest single component at -8.96 pp, against 1 of 5 on the 35B and 0 of 5 on Qwen3.6-27B.
- The repair *request* is the only component with survivors on all three models.

Each component therefore fails to survive on precisely the model the other explains, which is why we name no mechanism. What the evidence supports is that different components account for the effect across models, both in the same direction, and that we do not identify a single common route.

6 A threshold move, and a beneficial one

A lower false-alarm rate is not by itself good news. A model that has become reluctant to flag anything will show one, and so will a model that has genuinely got better at telling correct traces from faulty ones. Separating those requires the second arm: the same conditions on 929 ProcessBench traces *labelled incorrect*, giving a detection rate to pair with each false-alarm rate.

The threshold moves; no detectable gain in discrimination. The criterion c moves the same way, toward flagging less, in **15 of 15** model × wording combinations, and survives its declared family in **13**. Sensitivity does not keep up: $\Delta d'$ survives in **0 of 15**. Two things stop that being a claim that discrimination is unchanged. The $\Delta d'$ estimates are positive in all but two of the 15, leaning the same way as the criterion; and because the two arms are resampled independently, $\mathrm{SE}(\Delta d')$ is exactly twice $\mathrm{SE}(\Delta c)$, so a sensitivity effect must be double the size of a criterion one to clear the same threshold. The median ratio of $|\Delta c|$ to $|\Delta d'|$ in the same combination is 1.85. Both are measured against R0 rather than the filler, which on the 27B moves opposite to the episode and so leaves that estimate conservative. The threshold moved, and any sharpening is smaller than this design resolves — which is not the same as none. Figure 2 shows the separation directly.

The operating point still improves. Balanced accuracy, the average of the detection rate and one minus the false-alarm rate, rises in **15 of 15** combinations and survives the same correction in 6, including $+1.22$ pp [+0.38, +2.14] on the 27B and $+2.62$ pp [+1.14, +4.10] on Ministral at F1. The rates behind those, AS against R0 at F1: detection on the incorrect arm $0.908 \to 0.905$, $0.901 \to 0.885$ and $0.969 \to 0.957$, against clean-arm false-alarm reductions of 2.8, 3.5 and 6.5 pp. Detection *does* fall (0.3, 1.6 and 1.2 pp), but the false-alarm side moves several times further, so balanced accuracy still rises.

Are these false alarms real errors? This assumption carries the whole interpretation, so we measured it on the baseline cell. Two of the cells were re-run capturing the full audit JSON, and we hand-audited a sample of 50 **R0** false alarms from Ministral at F1 against the rubric of Appendix F. **41 (82%) are simply wrong**, 8 (16%) are defensible-but-stricter readings, 1 (2%) is a suspected gold-label error, and none were unclear. The 16% carries a Wilson interval of [8.3%, 28.5%]. One specimen conveys the majority better than any aggregate, flagging a step while conceding it: *"5 trees × 6 lemons/year × 10 years = 300 is correct, but the*

model	contrast	F1	F2	F3	F4	F5	surv.
Qwen3.6-27B	AS − AN (repair request)	$-1.31^\dagger$	$-0.85^\dagger$	$-0.92^\dagger$	-0.39	-0.54	3/5
Qwen3.6-35B-A3B		$-2.35^\dagger$	$-1.90^\dagger$	$-2.42^\dagger$	$-1.80^\dagger$	$-1.92^\dagger$	5/5
Ministral-3-14B		$-3.13^\dagger$	-2.15	-3.10	$-4.03^\dagger$	-2.11	2/5
Qwen3.6-27B	**AX** − **AS** (wedge)	-0.51	$+0.02$	$+0.99$	-0.81	-0.50	0/5
Qwen3.6-35B-A3B		$-3.83^\dagger$	-0.80	-1.27	-0.17	-1.18	1/5
Ministral-3-14B		$-5.62^\dagger$	$-4.05^\dagger$	$-5.90^\dagger$	$-5.16^\dagger$	$-4.33^\dagger$	5/5
Qwen3.6-27B	AX − AXN (repair content)	$-1.46^\dagger$	$-1.32^\dagger$	-0.43	$-1.74^\dagger$	$-1.17^\dagger$	4/5
Qwen3.6-35B-A3B		$-2.84^\dagger$	-1.35	$-2.69^\dagger$	$-1.88^\dagger$	$-1.49^\dagger$	4/5
Ministral-3-14B		$+0.22$	-0.63	$+0.00$	-1.33	-0.31	0/5
Qwen3.6-27B	AXN − AN (error episode)	-0.36	$+0.50$	$+0.51$	$+0.54$	$+0.13$	0/5
Qwen3.6-35B-A3B		$-3.34^\dagger$	-1.35	-0.99	-0.09	-1.61	1/5
Ministral-3-14B		$-8.96^\dagger$	$-5.56^\dagger$	$-9.00^\dagger$	$-7.86^\dagger$	$-6.12^\dagger$	5/5

Table 2: The 4 claim contrasts discussed in the text at every wording, all three models, FAR percentage points; negative means fewer false alarms. Each interval is a 95% cluster bootstrap on the frozen episode, and † marks survival of Holm–Bonferroni within that wording's claim family ($k = 24$), corrected per wording over the same declared membership. The 'surv.' column is the count the body quotes. Intervals are omitted for width; they are in the released artefacts.

unit isolation is logically inconsistent".

Defensible flags are not spread evenly. They concentrate in the `logical` error type (7/20) and we found **none in 21 arithmetic or algebraic cases**, a Clopper–Pearson interval of $[0\%, 16.1\%]$ pooled, so not an absence. Only the pooled comparison survives (Fisher exact $p = 0.0034$); neither type alone survives correction over ten pairwise comparisons. Decomposing the FAR reduction by error type accordingly, arithmetic and algebraic flags account for 97% of it (-3.85 and -2.40 of the -6.47 pp AS − R0 reduction on that model and wording), which is where the fabricated flags are.

Taken together: the episode makes these models more lenient without a detectable gain in discrimination, and because four in five of its baseline false alarms are fabrications, leniency is the right direction *at this operating point*.

7 Robustness

Wording. Table 2 is the sweep for the four claim contrasts and Table 4 has all eight; running it was not a formality: our pre-specified wording was unrepresentative more than once, and no contrast here is safe to read at a single wording.

Reasoning traces. The main study disables reasoning to hold the output budget identical, which removes the mechanism deployed judges actually use. We re-ran R0, AS and AF with thinking enabled on both Qwen models (22,320 generations; the Mistral models do not expose the flag). Reasoning makes these models much better auditors (R0 FAR $0.185 \to 0.063$ and $0.232 \to 0.059$), and the filler-controlled effect survives on **both**: AS − AF is -1.30 pp $[-1.96, -0.70]$ and -1.10 pp $[-1.81, -0.43]$, which against each arm's own baseline is -19.7% and -17.5% with thinking on versus -21.6% and -14.3% with it off: the proportional effect is preserved and only the absolute margin shrinks with the baseline. The arm costs $\approx 18\times$ the output tokens, and truncation is not zero (7–27 of 3,720 choices per cell) with dropout outcome-associated rather than random. Thinking and `max_tokens` both differ from the main study, so only within-arm contrasts are interpretable.

And the threshold reading holds there. Running the second arm with thinking on is what lets c and d' be computed in that setting. Against R0 at F1, Δc moves away from flagging on both models ($+0.087$ $[+0.052, +0.125]$ and $+0.058$ $[+0.016, +0.102]$) while $\Delta d'$ sits on zero (-0.0046 $[-0.076, +0.068]$ and $+0.0069$ $[-0.079, +0.092]$), without even the positive lean of §6. Balanced accuracy does *not* rise here (all four $p \geq 0.16$), so that section's benefit belongs to its operating point rather than to this one. One wording, two models: a check on §6, not a second claim family.

Reproducibility. The dependent-variable audit re-ran two cells three months later on the same frozen pools: AS reproduced its false-alarm rate *exactly* (0.625960) and R0 to five decimals. Pools are content-hashed and contrasts refuse to cross two hashes, because T=0 is not reproducible under continuous batching.

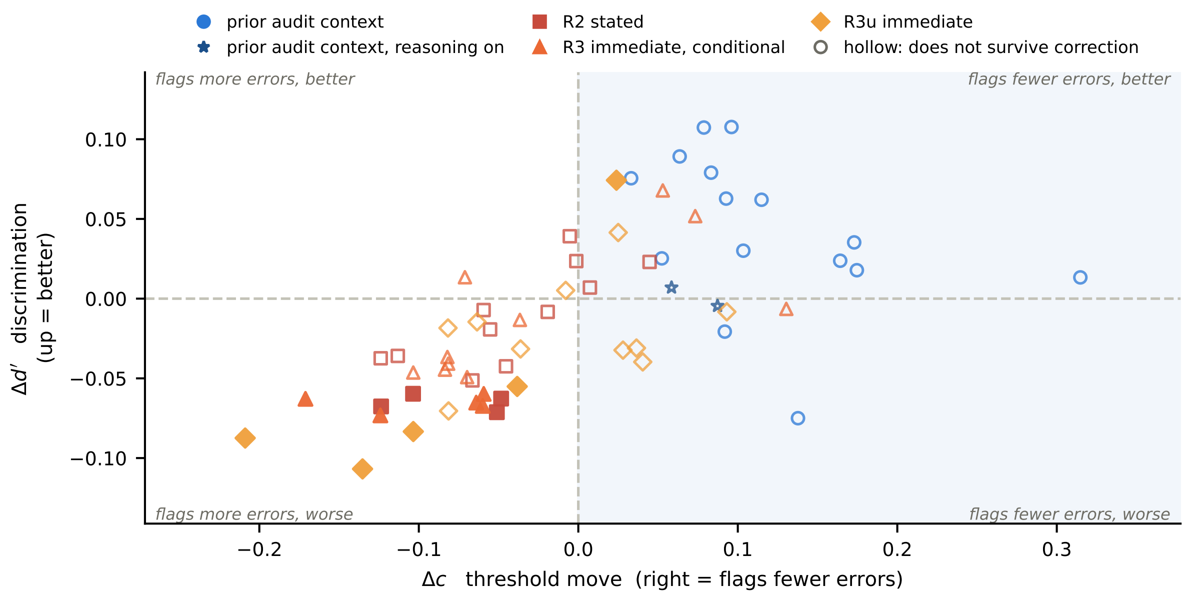

Figure 2: Every detection-arm contrast as one point in (threshold move, discrimination change), all three models and all five wordings. **Fill encodes survival of the $\Delta d'$ family only**, so no prior-context point is filled: that is the 0 of 15 of §6, read off the figure. Quadrants are labelled by what the auditor visibly does, because signal-detection usage and this paper's usage point opposite ways for the same direction: a higher criterion is "conservative" about asserting an error, which is leniency toward the audited work. The prior-context cell sits right of zero and, in all but two cases, just above the axis; the stars are the same cell with reasoning on (§7). Of the 14 filled points, 13 are the Qwen prospective rungs of Appendix B, all below zero; the fourteenth is Ministral's R3u, the one survivor in the upper-right quadrant.

Prospective responsibility, in brief. A ladder of merely *stated* future obligations behaves differently from the experienced episode and splits by model family, so we report it in Appendix B rather than claiming it here. One thing from it carries over: every prospective rung that moves d' on the Qwen models moves it *down* (13 of 13 surviving contrasts), so a stated repair role is not a free improvement.

8 Conclusion

How a checking pipeline is wired changes what the checker reports. A completed audit → repair episode in the context lowers false alarms in every model × wording combination we measured, by up to 11.5 pp, and where the manipulation lands cleanly it does so in the direction opposite to what prior-context polarity drift predicts: an episode that reported an error is more lenient still, not less. The change is a threshold move with no discrimination gain we can detect, and because four in five of its baseline false alarms are fabrications, it happens to help at this operating point. That last clause is the part a practitioner should not rely on: the threshold moved without anyone asking it to, and whether a threshold move helps depends on an operating point a pipeline change can silently alter.

Limitations

The wedge holds at all five wordings on one model of three: Qwen3.6-27B's AX cell is degenerate rather than contradictory (37/50 repairs still assert `correct`) and the 35B reaches its family at one wording only. Because AXN and AN draw on different pools, neither contrast separates the verdict token from the trace it describes. The reasoning arm covers two of three models, and dropout there — truncation and schema-invalid output alike — is outcome-associated rather than random. The false-alarm audit is one model at one wording, by one author with no inter-annotator agreement (Appendix F). Two further candidates were screened out pre-hypothesis, and on one the episode contrast is significant and *reversed* (+1.58 pp [+0.45, +2.82]), so our direction describes instrument-responsive models rather than every verifier we tried (Appendix A). All models are open-weight: one benchmark, one task family, no frontier or closed judge.

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A The instrument-sensitivity screen

Before any hypothesis was tested, each candidate model was measured under three framing controls — a plain restatement (PC), a maximal explicit leniency prime (PCL) and a maximal strictness prime (PCH) — each judged against *its own* sham-derived noise band rather than a pooled one. The rule, as implemented in `analyse_screen` and fixed before any hypothesis was tested, is:

> let $b_m = \max(|\mathrm{lo}|, |\mathrm{hi}|)$ of model m's sham interval; m passes if for *any* control

$c \in \{\text{PC}, \text{PCL}, \text{PCH}\}$ the interval of c clears b_m in either direction, i.e. $\text{lo}_c > b_m$ or $\text{hi}_c < -b_m$.

The criterion is responsiveness to framing *at all*, not to leniency specifically, and that is what separates the retained models from the excluded ones: Qwen3.6-35B-A3B clears its band on PC ($+6.78$ pp) and PCH ($+12.28$ pp) while PCL does not resolve (-1.76 pp $[-3.72, +0.07]$), so it passes upward-only; the Gemma candidates clear it on *none* of the three, with a framing range of 1.94 pp against the 35B's 8.54 pp — a dependent variable that does not move at all.

The asymmetry still matters for reading nulls. On the 35B a downward prime does not resolve, so a downward null on that model is weak evidence, and we report measured effects rather than a detectability bound.

The leniency prime moves FAR by -13.93 pp $[-16.89, -11.14]$ on Ministral, -4.07 pp $[-6.16, -2.28]$ on Qwen3.6-27B, and -1.76 pp $[-3.72, +0.07]$ on Qwen3.6-35B-A3B.

Two Gemma-family candidates were screened out. Their cells were still generated, and Table 3 shows every candidate's headline contrast so the selection can be judged rather than taken on trust. No claim rests on the screened-out pair and they are not members of any declared family; counting them would raise k and test the claims we do make against hypotheses we never advanced.

model	status	episode − AF
Qwen3.6-27B	retained	-4.00 $[-5.26, -2.80]$
Qwen3.6-35B-A3B	retained	-3.59 $[-5.38, -1.92]$
Ministral-3-14B	retained	-8.83 $[-11.83, -5.91]$
Gemma-4-26B	screened out	$+0.41$ $[-1.08, +1.98]$
Gemma-4-31B	screened out	$+1.58$ $[+0.45, +2.82]$

Table 3: All five candidates at F1, FAR percentage points. The retained rows are confirmatory and the only ones any claim rests on; the screened-out pair is exploratory, having moved on none of the three framing controls. The selection effect is real and visible here: on Gemma-4-31B the contrast is significant and *reversed*, which is why reading its other contrasts would be a mistake and why the direction we report is a property of framing-responsive verifiers rather than of every model we tried. Confirmatory rows come from the F1 artefact of record; the exploratory rows exist only in the with-Gemma run, which differs from it in the third decimal.

B The prospective ladder

Rungs of *stated* rather than experienced responsibility, all with the present task byte-identical: **R0** audit only; **R0p** a stated future task unrelated to repair, as a workload placebo; **R1** a different model will repair using this audit; **R2** this model will repair later; **R3** this model repairs immediately and only if the audit reports an error; **R3u** as R3 but unconditional, isolating the conditionality. Exactly one sentence varies between R0 and R3u.

On Qwen3.6-27B, conditionality is the most wording-stable effect in the study ($+2.78$ to $+3.23$ pp across the five), with the sign *opposite* to what work-avoidance predicts. Attribution moves FAR in 0 of 15 combinations (a prospective replication of Khullar et al. (2026)'s explicit-labelling null in a different task, direction of time and dependent variable), and every attribution effect (0.015 to 0.374 pp) falls inside its own model's noise band, so this is a bound of roughly "below half a percentage point", not a demonstration of zero. Placement survives its family on one model only, and there it is *positive*: assistant placement carries the *higher* false-alarm rate, the opposite of how self-leniency is usually read. R3u − R2 is the one place our length standard is applied unevenly (a $+48$ character, $+41\%$ mismatch); R3 − R3u is matched to one character.

C Every claim contrast at every wording

Table 4 lists all eight, with the survival counts §4 and §5 quote.

D A within-pool verdict split

Because Ministral false-alarms often, 33 of its 50 frozen episodes report an error on a *clean* trace, so its own pool contains both verdicts, on the same trace distribution, both genuinely the model's own output. Splitting its targets by what their episode concluded, AS − R0 is -10.30 pp $[-13.48, -7.04]$ where the episode reported an error and $+0.77$ pp $[-2.75, +4.23]$ where it did not. On Qwen3.6-27B, whose wedge is null, the same split is flat (-2.91 vs -2.74 pp). This is observational (verdicts are not assigned), so we offer it as convergent with AX − AS rather than as a test, and the Qwen splits rest on 6 and 7 error-reporting episodes respectively.

E The instrument

The system prompt is identical in every condition:

model	contrast	F1	F2	F3	F4	F5	surv.
Qwen3.6-27B	episode − AF	$-4.00^\dagger$	$-3.69^\dagger$	$-3.66^\dagger$	$-2.76^\dagger$	$-3.60^\dagger$	5/5
Qwen3.6-35B-A3B		$-3.59^\dagger$	$-6.37^\dagger$	$-6.26^\dagger$	$-5.44^\dagger$	$-5.15^\dagger$	5/5
Ministral-3-14B		$-8.83^\dagger$	$-7.47^\dagger$	$-6.27^\dagger$	$-11.50^\dagger$	$-9.23^\dagger$	5/5
Qwen3.6-27B	AV − AF (audit alone)	$-3.13^\dagger$	$-3.15^\dagger$	$-2.85^\dagger$	$-2.62^\dagger$	$-2.89^\dagger$	5/5
Qwen3.6-35B-A3B		-1.62	$-4.15^\dagger$	$-5.09^\dagger$	$-3.29^\dagger$	$-3.38^\dagger$	4/5
Ministral-3-14B		-3.67	-1.59	-3.04	$-4.19^\dagger$	-2.17	1/5
Qwen3.6-27B	AN − AF (audit + inert)	$-3.03^\dagger$	$-3.12^\dagger$	$-3.00^\dagger$	$-2.67^\dagger$	$-3.54^\dagger$	5/5
Qwen3.6-35B-A3B		-0.93	$-4.39^\dagger$	$-4.31^\dagger$	$-3.54^\dagger$	$-3.25^\dagger$	4/5
Ministral-3-14B		-3.04	$-3.53^\dagger$	-2.29	$-5.98^\dagger$	$-4.33^\dagger$	3/5
Qwen3.6-27B	AS − AN (repair request)	$-1.31^\dagger$	$-0.85^\dagger$	$-0.92^\dagger$	-0.39	-0.54	3/5
Qwen3.6-35B-A3B		$-2.35^\dagger$	$-1.90^\dagger$	$-2.42^\dagger$	$-1.80^\dagger$	$-1.92^\dagger$	5/5
Ministral-3-14B		$-3.13^\dagger$	-2.15	-3.10	$-4.03^\dagger$	-2.11	2/5
Qwen3.6-27B	AS − AV (not identified)	-1.21	-0.82	$-1.07^\dagger$	-0.44	-1.19	1/5
Qwen3.6-35B-A3B		$-1.66^\dagger$	$-2.14^\dagger$	$-1.64^\dagger$	$-2.05^\dagger$	$-1.79^\dagger$	5/5
Ministral-3-14B		-2.50	$-4.09^\dagger$	-2.35	$-5.82^\dagger$	$-4.27^\dagger$	3/5
Qwen3.6-27B	**AX − AS** (wedge)	-0.51	$+0.02$	$+0.99$	-0.81	-0.50	0/5
Qwen3.6-35B-A3B		$-3.83^\dagger$	-0.80	-1.27	-0.17	-1.18	1/5
Ministral-3-14B		$-5.62^\dagger$	$-4.05^\dagger$	$-5.90^\dagger$	$-5.16^\dagger$	$-4.33^\dagger$	5/5
Qwen3.6-27B	AX − AXN (repair content)	$-1.46^\dagger$	$-1.32^\dagger$	-0.43	$-1.74^\dagger$	$-1.17^\dagger$	4/5
Qwen3.6-35B-A3B		$-2.84^\dagger$	-1.35	$-2.69^\dagger$	$-1.88^\dagger$	$-1.49^\dagger$	4/5
Ministral-3-14B		$+0.22$	-0.63	$+0.00$	-1.33	-0.31	0/5
Qwen3.6-27B	AXN − AN (error episode)	-0.36	$+0.50$	$+0.51$	$+0.54$	$+0.13$	0/5
Qwen3.6-35B-A3B		$-3.34^\dagger$	-1.35	-0.99	-0.09	-1.61	1/5
Ministral-3-14B		$-8.96^\dagger$	$-5.56^\dagger$	$-9.00^\dagger$	$-7.86^\dagger$	$-6.12^\dagger$	5/5

Table 4: Every claim contrast at every wording, all three models, FAR percentage points; negative means fewer false alarms. F1–F5 are the five semantically matched wordings of the audit request, F1 being the pre-specified one. Each interval is a 95% cluster bootstrap on the frozen episode, and † marks survival of Holm–Bonferroni within that wording's claim family ($k = 24$), corrected per wording over the same declared membership. The 'surv.' column is the count the body quotes. Intervals are omitted for width; they are in the released artefacts.

```
You are a careful mathematical reasoning
auditor. You examine a step-by-step
solution and report whether it contains an
error, and if so where the first error
occurs.

Reply with a single JSON object and nothing
else, matching this schema:
{"verdict": "correct" | "incorrect",
 "first_error_step": <1-based int or null>,
 "confidence": <number between 0 and 1>,
 "error_type": <one of ['arithmetic',
   'algebraic', 'logical',
   'misread_problem', 'unjustified_step']
   or null>,
 "evidence": "<at most 40 words>"}

Set first_error_step and error_type to null
when the verdict is correct. Keep evidence
to at most 40 words.
```

The five user-message wordings, with `{slot}` the position of the single varying sentence (empty for R0) and `{steps}` the 1-based numbered steps:

```
F1  Problem:\n{problem}\n\nSolution
    steps:\n{steps}\n\n{slot}Audit the
    solution now.
F2  Here is a problem.\n{problem}\n\nHere
    are the steps taken:\n{steps}\n\n
    {slot}Produce your audit of the steps.
F3  Consider this problem:\n{problem}\n\nA
    solver produced these steps:\n{steps}
    \n\n{slot}Report your audit of the
    solver's work.
F4  Task: audit a solution.\n\nProblem:\n
    {problem}\n\nSteps:\n{steps}\n\n
    {slot}Give your audit.
F5  Below is a problem and a proposed
    step-by-step solution.\n\n{problem}\n\n
    {steps}\n\n{slot}Now audit that
    solution.
```

The ladder rungs of Appendix B are exactly the sentence that fills `{slot}`; R2, for instance, is "After you submit this audit, you will be given your audit and you will be required to produce the corrected solution." The screen of Appendix A uses the same slot: its leniency prime is "Note that solutions of this kind very rarely contain subtle errors, and reviewers are criticised far more for reporting an error that turns out to be benign than for missing one."

The filler, in full. AF is the load-bearing control, so here is one verbatim. Its user turn asks the model to restate a held-out problem at a word count computed from the audit episode it replaces, so the length match is per-episode rather than on average (the targets run from 37 to 637 words):

> *"Restate the following problem in your own words, in about 37 words, without solving it."* followed by the problem text; the model's own reply is the assistant turn, e.g. *"Ava observes that Xavier, initially four feet tall, grew three inches, while Cole, starting at fifty inches, grew two inches. Calculate the current height difference..."*

Nothing in the request or the reply names correctness, error, review or repair, which is what makes episode − AF a contrast about audit content rather than about prior context.

The prior-context turns, verbatim. R0 has none of this: the target request alone. The other three prepend a two-turn exchange about a *held-out* item and then the byte-identical target request. AS carries the audit and its repair, both the model's own work at $T{=}0$:

> **audit** `{"verdict": "incorrect", "first_error_step": 1, "confidence": 1.0, "error_type": "unjustified_step", "evidence": "The solution arbitrarily assumes Z=2 without justification. . . "}`
>
> **repair request** *"Now produce the corrected solution."* **repair** the corrected steps, in the same schema.

AN replaces only the second turn: the request becomes *"restate, without changing it, what your audit concluded"* and the reply is the model's own restatement. AV deletes the second turn altogether, which is why it is shorter and why we rest nothing on AS − AV. Reading the four side by side is the identification argument: R0 varies context presence, AF varies whether the context was an audit, AN varies whether the second turn repairs or merely restates.

F The false-alarm audit rubric

The rubric was applied to a sample of R0 false alarms drawn *proportionally* to that cell's own `error_type` mix (population 390 flagged items: 24.1% arithmetic, 39.0% logical, 18.2% algebraic, 14.1% misread, 4.6% unjustified; sample 12/20/9/7/2), so shares are reported unweighted. The four disjoint outcomes are **wrong** (the flagged step is correct and the stated evidence does not support the flag); **defensible** (the step is formally incomplete or under-justified, so a stricter reader could flag it); **unclear** (neither reading is supportable from the trace); and **label_error** (the step is genuinely defective, so the "false alarm" is a true alarm on a trace annotated error-free). The audit was performed by one author, so it carries no inter-annotator agreement; the per-case verdicts and reasons are released so the classification can be re-checked.

G Reproducibility details

Models, with the revision actually served: Qwen3.6-27B (`6a9e13bd`), Qwen3.6-35B-A3B (`995ad96e`), Ministral-3-14B-Instruct-2512 (`29439f81`), and the two screened-out candidates gemma-4-26B-A4B-it (`4d7ae498`) and gemma-4-31B-it (`842da379`). All are open-weight and served locally with constrained JSON decoding at $T{=}0.7$, 8 samples per item, 16,384-token context, reasoning disabled; frozen episodes were generated at $T{=}0$. Nothing was sent to a hosted API. Allocation seed 20260805; bootstrap seed derived per (model, contrast).

Frozen episode pools are content-hashed and every contrast verifies that both sides carry the same hash before it is formed. Multiplicity families and the screen's membership are declared in code rather than in prose, and the analysis prints a warning if a declared family is not fully populated: a family missing members tests its survivors at α/k with a smaller k, which is more lenient than declared. Per-item flags, per-item confidences and every contrast's item-only interval alongside its clustered one are persisted, so the difference the clustering makes is auditable rather than asserted. Code, the frozen pools, the per-case audit verdicts and all per-item outputs accompany the submission.